

CLOUDFLARE RADAR • ANNUAL INTELLIGENCE REPORT

2024 Year in Review

A data-driven analysis of global Internet traffic, technology adoption, connectivity disruptions, and security threats observed through Cloudflare's network in 2024.

- 17% Traffic Growth
- 225 Disruptions
- Post-Quantum Encryption
- Security Threats
- AI Services Rise

2024.



17% Global Traffic Growth and Starlink's 3.3x Surge

YOY TRAFFIC GROWTH

+17%

Global Internet traffic growth in 2024, driven by commerce, communication, entertainment, and transportation reliance.

STARLINK GROWTH

3.3x

SpaceX Starlink traffic surge in 2024, expanding satellite connectivity to previously underserved locations worldwide.



Google Leads Top 10 Services as Generative AI Rises



Source: Cloudflare 1.1.1.1 DNS resolver aggregate data

Google's 2nd Consecutive Year at #1

Google maintained its position as the most popular Internet service globally, topping rankings for the second consecutive year based on DNS query volume.

Generative AI on the Rise

Generative AI services continued their rapid rise throughout 2024 while the Metaverse receded from prominence.

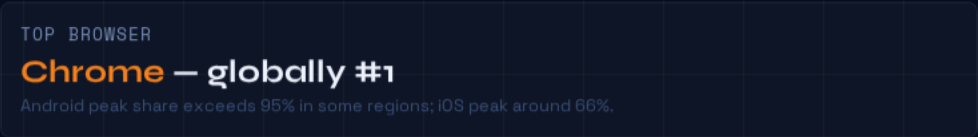
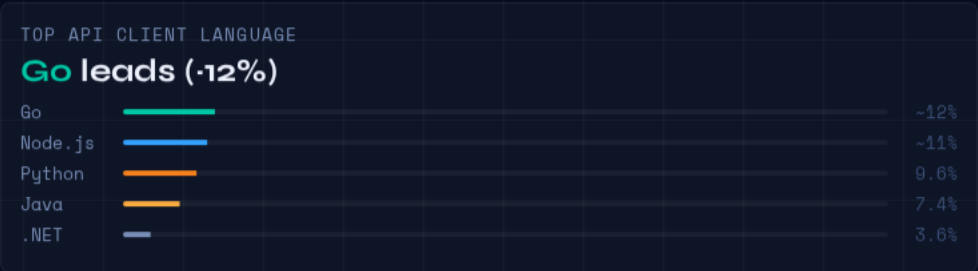
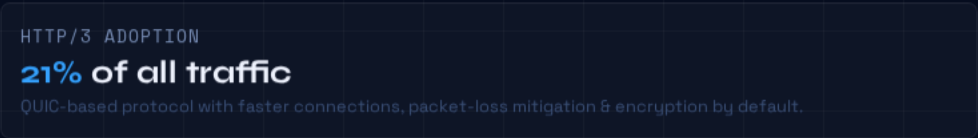
▲ Accelerating growth in 2024

Search Engine Share

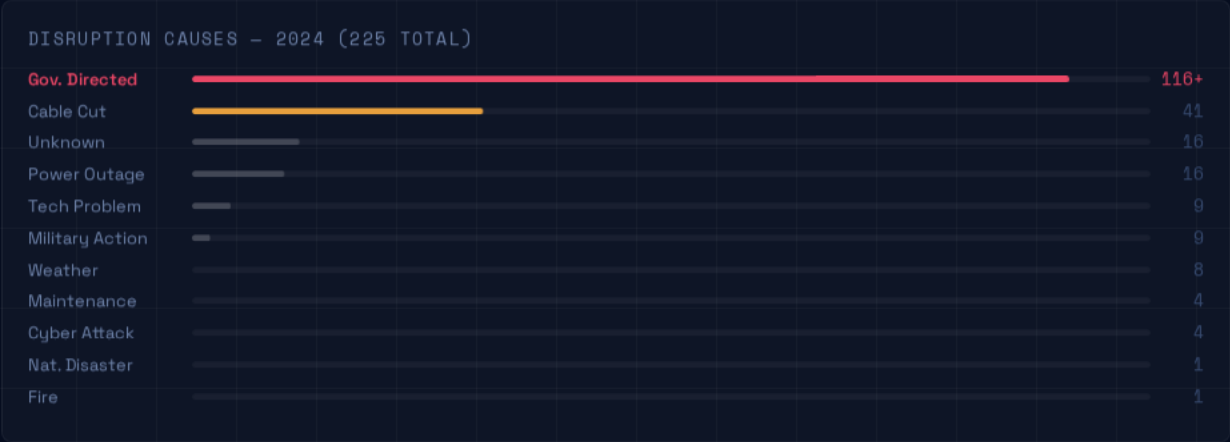
Google holds **89%** of search referral traffic. Yandex 3.1%, Baidu 2.7%, Bing 2.6%.



Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



225 Major Disruptions: Government Shutdowns Lead Outage Causes



116+

GOVERNMENT DIRECTED

Over half of all 225 disruptions in 2024 were intentional government-mandated shutdowns — more than all other causes combined.

Cable cuts (41) represent physical infrastructure failures — often difficult to predict or prevent.

Power outages (16) and unknown causes (16) reflect ongoing resilience challenges in global connectivity infrastructure.

Source: Cloudflare Radar Outage Center, tracking scope 5 duration via Cloudflare traffic data.

6.5% of Traffic Mitigated: Persistent Threats & Bot Concentration

TRAFFIC MITIGATION RATE

6.5%

Of all traffic processed by Cloudflare — indicating the persistent, large-scale nature of ongoing Internet threats.

MOST ATTACKED SECTOR

Gambling
& Games

The gaming and gambling industry was the top targeted sector by attackers in 2024.

BOT TRAFFIC CONCENTRATION

68%

Of global bot traffic originated from just the top 10 countries — highly concentrated geographic source.

LOG4J PERSISTENCE

CVE-2021-44228

First disclosed Dec 2021 — still actively exploited 3+ years later, underscoring how long critical vulns remain live targets.

EMAIL SECURITY THREATS

4.3%

of processed emails were malicious

43%

of malicious emails contained a deceptive link

.bar

TLD with largest share of malicious & spam email

Five Takeaways Shaping the 2025 Internet Outlook

01 Internet Traffic Growth Trajectory Continues

+17% YoY confirms the structural reliance on the Internet. Satellite (Starlink 3.3x) is bridging the digital divide for underserved regions.

02 Post-Quantum Encryption Momentum Is Irreversible

Chrome 124's default PQ enablement drove adoption from 2% → 13% of TLS 1.3 traffic. Browser rollouts will push adoption higher — prepare infrastructure now.

03 Connectivity Fragility from Government Actions

116+ of 225 disruptions were government-directed — over half of all outages. Resilient multi-path routing is a geopolitical necessity, not just a technical one.

04 Persistent Security Threats Require Layered Defense

6.5% mitigation rate and ongoing Log4j exploitation (3+ years post-disclosure) confirm that legacy vulnerabilities remain live attack surfaces indefinitely.

05 Evolving Developer Ecosystem & AI Services

Go tops API client languages at ~12%. Generative AI services surged in traffic rankings while the Metaverse faded — signaling where developer and user attention is shifting.

